UNIT 2 · 8 hours

Time Response of Second-Order Systems



Time Domain Analysis of Control Systems

Control Systems (22EEG53)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Write the standard second-order transfer function
- Classify the response: undamped, underdamped, critically damped, overdamped
- Write and sketch the step response in each case
- Find ωn and ζ from a given transfer function

Syllabus context

Time domain analysis of control systems: standard test signals, time response of zero, first and second order systems, time domain specifications, type of systems, static error constants and steady-state error, generalized error series.

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Standard second-order system

$$C/R = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

standard form

$$s = -\zeta\omega_n \pm \omega_n \sqrt{(\zeta^2 - 1)}$$

closed-loop poles

- ωn = undamped natural frequency (rad/s); ζ = damping ratio.
- Servo example: G = K/[s(s + a)] with unity feedback: ωn = √K and ζ = a/(2√K).
- Unit step input: R(s) = 1/s.

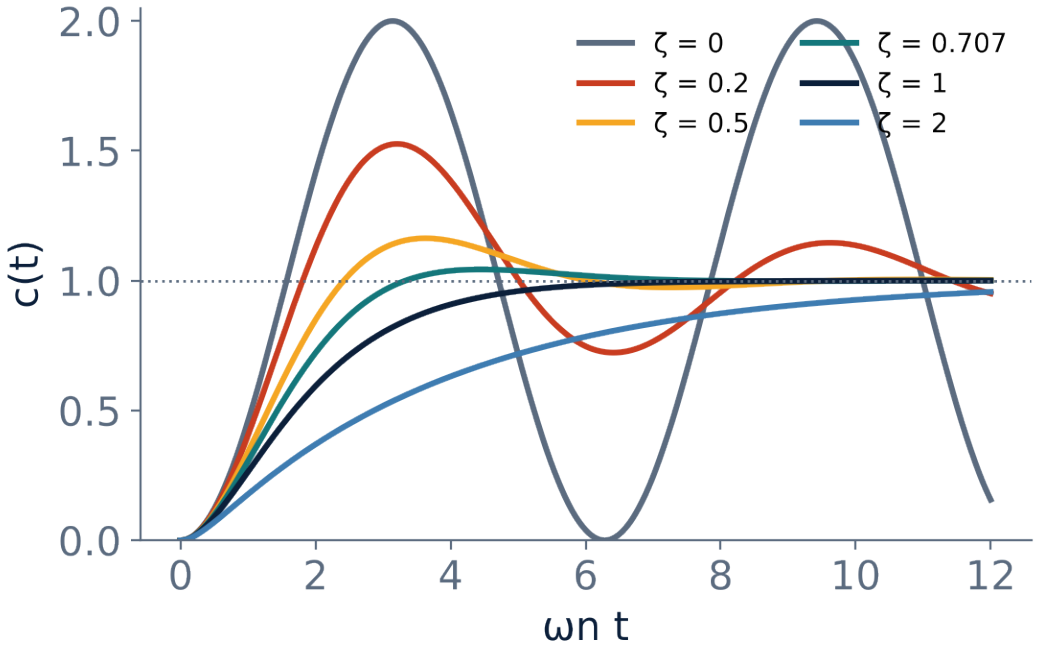
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Four damping cases

ζ	Poles	Type	Step response
0	±jωn	Undamped	Sustained oscillation, 1 − cos ωn t
0 < ζ < 1	−ζωn ± jωd	Underdamped	Decaying oscillation with overshoot
1	−ωn (double)	Critically damped	Fastest without overshoot
> 1	Two real negative	Overdamped	Slow, no overshoot

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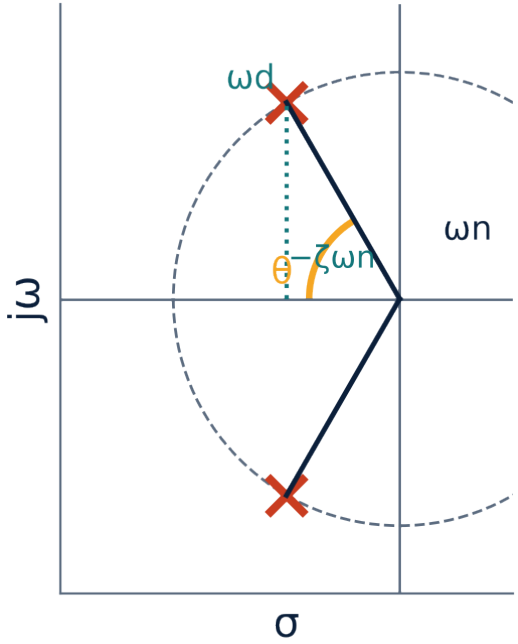
Step responses for different ζ



- ζ = 0: undamped, never settles.
- Small ζ: large overshoot, many oscillations.
- ζ ≈ 0.7: fast, small overshoot (≈ 4 %).
- ζ = 1: critically damped.
- ζ > 1: sluggish.

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Pole location and ζ



- Poles at distance ωn from the origin.
- cos θ = ζ where θ is measured from the negative real axis.
- Real part −ζωn gives decay rate; imaginary part ωd gives oscillation.
- Constant ζ ⇒ radial line; constant ωn ⇒ circle.

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Underdamped step response

$$c(t) = 1 - [e^{(-\zeta\omega_n t)}/\sqrt{(1-\zeta^2)}] \sin(\omega_d t + \varphi)$$

0 < ζ < 1

$$\varphi = \cos^{-1} \zeta$$

phase angle

$$\text{Critical: } c(t) = 1 - (1 + \omega_n t) e^{(-\omega_n t)}$$

ζ = 1

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Numerical: classify the response

Problem Find ω_n, ζ and the type for (a) $25/(s^2+6s+25)$, (b) $4/(s^2+4s+4)$, (c) $9/(s^2+9s+9)$.

- (a) $\omega_n = 5$, $2\zeta\omega_n = 6 \Rightarrow \zeta = 0.6$: underdamped
- (b) $\omega_n = 2$, $2\zeta\omega_n = 4 \Rightarrow \zeta = 1$: critically damped (double pole at -2)
- (c) $\omega_n = 3$, $2\zeta\omega_n = 9 \Rightarrow \zeta = 1.5$: overdamped (poles -1.15, -7.85)

Answer: (a) $\zeta = 0.6$ (b) $\zeta = 1$ (c) $\zeta = 1.5$

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Numerical: servo gain and damping

Problem A unity-feedback system has $G(s) = K/[s(s+4)]$. Find K for $\zeta = 0.6$ and the resulting ωn.

- Characteristic equation: $s^2 + 4s + K = 0 \Rightarrow \omega n^2 = K,\ 2\zeta\omega n = 4$
- $\omega n = 4/(2 \times 0.6) = 3.33$ rad/s
- $K = \omega n^2 = 11.1$

Answer: $K = 11.1$, $\omega n = 3.33$ rad/s

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Key takeaways

- Standard form $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$
- $\zeta = 0$, $0<\zeta<1$, $\zeta = 1$, $\zeta > 1$: undamped, under, critical, overdamped
- $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$; $\cos\theta = \zeta$
- For $G = K/[s(s+a)]$: $\omega_n = \sqrt{K}$, $\zeta = a/(2\sqrt{K})$

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Practice questions

Q1. Write the step response for $0 < \zeta < 1$ and sketch it.

Q2. Find ωn and ζ for $16/(s^2 + 4s + 16)$.

Q3. $G = K/[s(s + 2)]$: find K for critical damping.

Q4. Locate the poles for $\zeta = 0.5$, $\omega n = 10$.